# Data-Driven Analysis and Forecasting of COVID-19 Third Wave in SAARC Countries

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***Abstract*—COVID-19 is a global pandemic that has affected millions of people worldwide. Effective forecasting of the spread of COVID-19 is crucial for proactive planning, public health interventions, adopting vaccination strategies, and raising public awareness. It enables a more effective and coordinated response to the pandemic, ultimately saving lives and reducing the impact on societies. Like many other parts of the globe, the eight countries under the 'South Asian Association for Regional Cooperation' (SAARC) have faced different challenges in dealing with the repeated waves of this pandemic. In this paper, we have analyzed the aftermath of COVID-19's third wave in SAARC countries and proposed a model for forecasting future situations using Machine Learning (ML) algorithms. We have used data from various sources to visualize and compare the cases, tests, deaths, and vaccinations in each of the member countries. We provided a thorough analysis of different aspects of the COVID-19 third wave across different countries and compared the performance of ML algorithms like Linear Regression, Polynomial Regression, KNN Regression, and Time Series Forecasting to make effective predictions utilizing two years of data. Our findings show that COVID-19 has affected SAARC countries in different ways, depending on their population, healthcare system, pandemic management, and vaccination rate. This study can be helpful in the future to forecast the severity of virus spreading based on statistical data, which is crucial to adopting effective strategies.**

***Keywords*—Predictive Systems, Coronavirus, Pandemic, Time Series Forecasting**

## I. Introduction

COVID-19 is a novel coronavirus disease that emerged in China in late 2019 and spread to almost every country, causing a global health crisis [1]. The virus is highly contagious and can cause mild respiratory infections to severe neurological complications [2]. The disease poses a serious challenge to the human immune system, as no specific treatment or vaccine is available yet to completely get rid of the virus [3]. Therefore, it is crucial to understand the epidemiology, transmission, and prevention of COVID-19 and to develop effective methods for predicting along with controlling its spread.

This global pandemic has affected all SAARC countries, where India has the highest number of cases and deaths, followed by Bangladesh and Pakistan. India's large population has contributed to the fast transmission of the virus. However, India also has a high recovery rate, along with Bangladesh and Pakistan [4]. The existing literature on COVID-19 has focused on its origin, symptoms, diagnosis, immunity, and impact [5]. Machine Learning (ML) based techniques can be utilized as vital to analyze and forecast the COVID-19 cases and deaths in different regions and countries [6].

In this work, we have made an attempt to fight the coronavirus by using data analysis to predict the future trend of the virus in SAARC countries. This can help in understanding the problem and raise awareness. We have used data visualization, data-driven algorithms, and solid predictions to present our research work. We have also discussed how this work can be improved and what future efforts are needed.

## II. Literature Review

Saleem *et al.* studied the impact of COVID-19 on the stock market performance and volatility in some SAARC countries [7]. They found that the virus outbreak had negative effects on the stock market and that the lack of preparedness and preventive measures resulted in significant economic losses. Kumar *et al.* added to the previous research and showed the harmful effects of viral infections on the stock market [8]. Dananjayan *et al.* discussed the role of Artificial Intelligence (AI) in combating the pandemic [9].

The work of Agarwal and Bansal [10] used ML to predict COVID-19 cases in South Asia and found exponential growth in some countries. Another work used Neural Networks to forecast COVID-19 cases and death rate in Hungary [11]. Prakash *et al.* used the ML technique to analyze the impact of COVID-19 on different age groups and found that Random Forest models performed better [12]. Mantaro *et al.* used Support Vector Regression (SVR) and Susceptible-Infected-Recovered (SIR) Models to predict the end of the COVID-19 pandemic in Indonesia under two scenarios, best-case and worst-case, to assist the policymakers [13]. The best-case scenario assumed the current daily case as the maximum, while the worst-case scenario assumed the maximum case of India. SVR predicted the end of the pandemic in January 2021 for the best-case scenario and in March 2021 for the worst-case scenario. SIR Model predicted the end of the pandemic in January 2021 for both scenarios but with a huge difference in the number of infected people (450,000 vs. 5,500,000).

Liu and Xiao used polynomial regression to predict and analyze the number of COVID-19 cases in Xinjiang in July 2020, based on the daily data and a transmission model

[14]. They compared the national and global outbreaks and highlighted the importance of prevention, isolation, and control measures. They suggested that more research and data are needed to improve the prediction model and account for asymptomatic cases. Yang and Chen used different regression models to simulate the pandemic trend in the US based on alternative data sources such as Twitter and proposed a novel method for estimating pandemic trends using state-level data [15]. A similar kind of analysis is conducted in [16], which concluded that despite linear regression outperformed fully connected neural networks with limited data, it might improve with more data. All the findings suggest that to fight COVID-19 in a timely way, the analysis and prediction utilizing the existing data are crucial.

## III. Methodology

To successfully conduct the research, at first, we collected data, performed some preprocessing steps, and then transformed the data into a consistent and uniform format. Afterward, we applied different data visualization techniques and evaluated the performance of different ML-based algorithms for effective forecasting.

### A. Data Collection and Dataset Preparation

This study is data-driven and relies on statistics related to the COVID-19 pandemic that were gathered from newspapers, websites, and other sources from December 31, 2019 to February 21, 2022. The data covers eight countries and focuses on four factors: cases, deaths, tests, and vaccinations. The collected raw data is cleaned, organized, and smoothed using different techniques to remove outliers, noise, and missing values. For example, we used the Savitzky–Golay filter, which is a digital filter that applies a polynomial to a small window of data points and adjusts them according to the least squares method. We adopted this data preprocessing step to enhance the quality and reliability of the data and to facilitate the subsequent analysis and interpretation of the results.

### B. Prediction Model

Four ML-based methods, namely, Linear Regression, Polynomial Regression, KNN Regression, and Time Series Forecasting, were applied to forecast the COVID conditions in SAARC nations based on the confirmed number of cases, deaths, tests, and vaccinations.

*1) Linear Regression:* Linear regression is a statistical method that models the relationship between a scalar outcome and one or more explanatory variables. We used multiple linear regression, which is suitable for more than one explanatory variable. The multiple linear regression function is:

$$Y_i = \beta_0 + \beta_1 x_1 + ... + \beta_n x_n + \epsilon_i \tag{1}$$

where $Y_i$ is the outcome, $x_i$ are the explanatory variables, $\beta_i$ are the coefficients, and $\epsilon_i$ is the intercept. We used multiple linear regression to find the association between the outcome and the explanatory variables.

*2) Polynomial Regression:* We used polynomial regression, which is a type of multiple linear regression that models the relationship as an $n^{th}$ degree polynomial. This can solve the underfitting problem by increasing the degree of the polynomial. The regression function is:

$$Y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2^2 + \beta_3 x_3^3 + ... + \beta_n x_n^n \tag{2}$$

where $Y_i$ is the dependent and $x_i$ is the independent variable, and $\beta_i$ are the coefficients. We have chosen the optimal degree by comparing the performance score and variance of our model. We also handled outliers and overfitting by using regularization techniques to penalize large coefficients.

*3) KNN Regression:* We also used the KNN regression method, which is a non-parametric technique that predicts the outcome by averaging the nearest neighbors in the feature space. The KNN regression function is:

$$y(x) = \frac{1}{k} \sum x_i \epsilon N(x) y_i \tag{3}$$

where $N(x)$ is the set of nearest neighbors, and $k$ is the number of neighbors. The KNN method can also classify by taking the majority vote of the neighbors. We chose $k$ by cross-validation and found that KNN regression worked well and agreed with previous studies.

*4) Time Series Forecasting:* Finally, we used time series forecasting, which is a scientific method that predicts future events based on past data that has a temporal component. This technique uses a linear model to relate the response variable at time $t$ to one or more predictor variables at time $t$ or previous timestamps. The formula for a regression method of time series forecasting is:

$$y_t = X_t \beta + \epsilon_t \tag{4}$$

where $y_t$ is the response variable at time $t$, $X_t$ is a design matrix of predictor variables at time $t$ or previous times, $\beta$ is a vector of regression coefficients, and $\epsilon_t$ is a random error term at time $t$.

## IV. Results and Discussion

At first, we examined the situation in all SAARC countries through data visualization, as mentioned in Figure 1. For all nations, the first day is December 31, 2019, and the final day is February 21, 2022.

### A. Cases

COVID-19 growth is similar for all SAARC nations, and the cases have spiked in three waves. Most nations had the first wave in August and September of 2020. Bhutan, the Maldives, and Sri Lanka controlled it well. The second wave started around March and April 2021. It hit India the worst, with cases rising from 1 million to 3 million in 100 days. Bangladesh had a steady increase. Afghanistan had a serious situation due to unrest. Pakistan and Nepal had a similar rise. Sri Lanka's situation worsened at the end of the second wave, with cases jumping from 200,000 to 500,000 in 100 days.

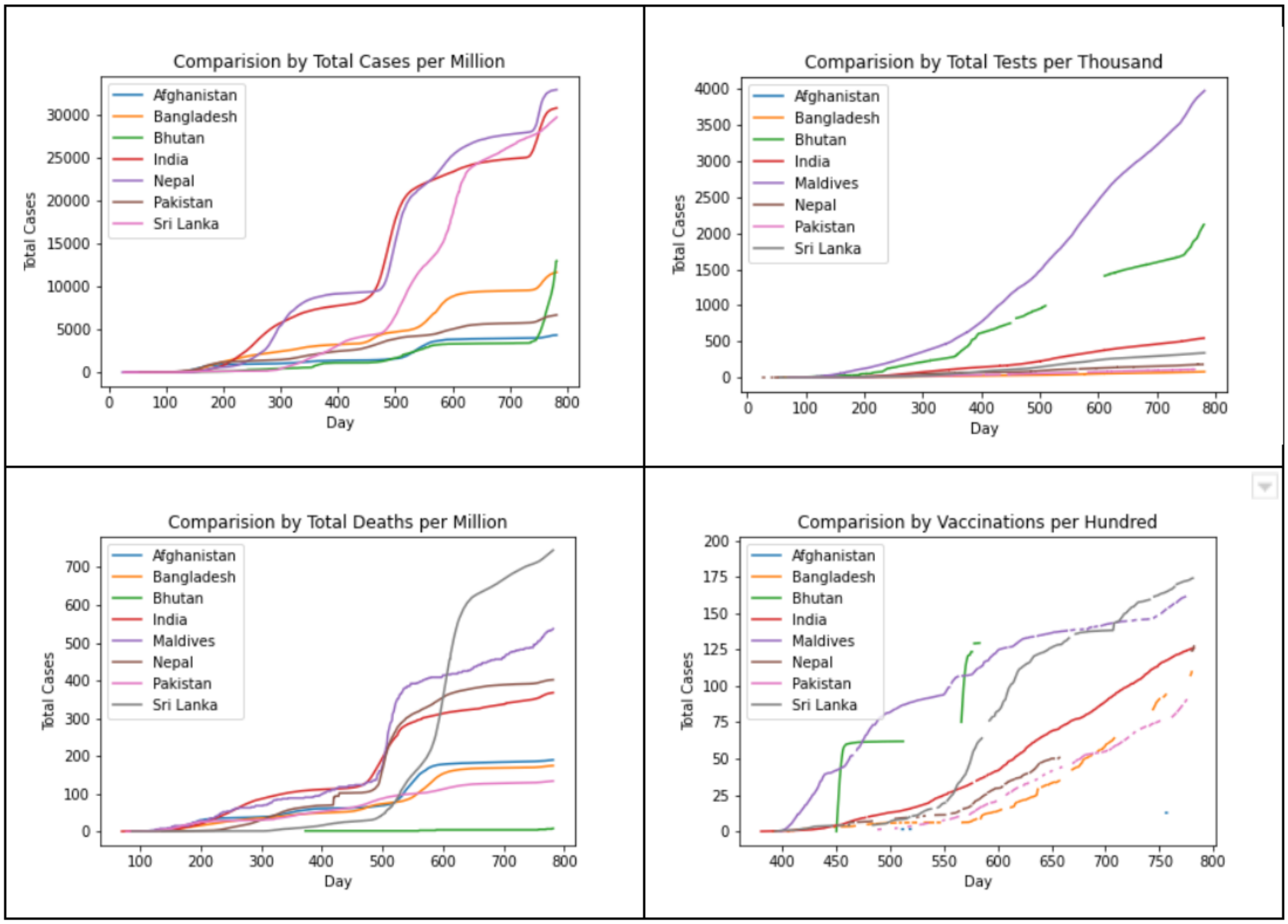


Fig. 1: Cases, Tests, Deaths and Vaccinations Comparison in SAARC Countries

The latest outbreak happened in January 2022, and it was the most infectious wave, as some countries that had limited the virus saw big increases, like the Maldives, Bhutan, and Sri Lanka. It is observed that Afghanistan, India, Bangladesh, Pakistan, and Nepal had fewer increases, as many of their people developed herd immunity from the 2nd wave [17].

### B. Tests

To measure how well a country is fighting a virus, we plotted the total number of tests against the number of days for each country. There were no data about the number of tests for Afghanistan. The graphs show that the number of tests per day is steady for all countries, so the total number of tests keeps growing. The white spaces show when no tests were done. These times are between the three waves when the new cases per day were low.

The Total Tests per Thousand figure shows that the Maldives did the best in this regard. Some people were tested up to three times to make sure they were not affected. Since the country is a popular tourist place, they needed to make sure that the place was safe for visitors. Following the Maldives, Bhutan also did well in testing as they are also a place of tourist attraction. After them are the five countries that did not do well in testing their people. India, Bangladesh, and Pakistan did poorly in this category due to their huge population, where Bangladesh has the lowest score.

### C. Deaths

Covid-19 has been one of the deadliest pandemics in human history, taking away around 6 million lives [18]. Many things affect this, like hospital beds, doctors, supply chains, and politics. Most SAARC nations have public and private healthcare, but most governments cannot help all the patients in a crisis. The death rate of most SAARC countries is similar. The first wave has a rise in death, then a bigger rise in the second wave, and a smaller rise in the third wave. The comparison graph shows the same pattern, with each country having a significant rise in deaths in the second wave. Sri Lanka had the most deaths per million but did well until the end of the second wave.

### D. Vaccination

The vaccination rate of a country shows how well the government is getting its people ready to fight the infection. SAARC nations mostly use AstraZeneca's Vaxzevria, Sinopharm's BBIBP-CorV, and Janssen's Ad26.COV 2.S, Pfizer BioNTech's Comirnaty, Sinovac's CoronaVac, and Moderna's Spikevax vaccines, based on data from UNICEF's Covid-19 vaccine market dashboard. Vaccination rates are higher in the Maldives and Sri Lanka than in the other countries. The Maldives started vaccinating people before the first wave, but Sri Lanka started vaccination during the second wave and increased its numbers a lot. Next, India has a steady vaccination rate. India and Pakistan had a rise in vaccination rates at the start and end of each wave. Bhutan, Nepal, and Bangladesh are behind in vaccination, with Afghanistan's war-torn country having the lowest vaccination rate.

### *E. Cases per Tests*

Figure 2 shows the number of affected cases per test for each country, which tells us when waves have started and ended in each country. The last graph shows the average number of cases per test for all eight countries together. This gives us a big picture of the COVID outbreak in the South Asian region. Covid-testing data was unavailable for Afghanistan, so the trend could not be analyzed. Bangladesh, Bhutan, India, Maldives, Nepal, and Sri Lanka experienced three waves, reaching their peaks in mid-2020, mid-2021, and early 2022 respectively. Pakistan had only one noticeable wave in mid-2020, and the data became unreliable. India and Maldives suffered the most from the second wave, which was very severe and prolonged. Sri Lanka faced the most challenges from the third wave, which started late and lasted long.

The combined average shows a similar picture; we can see that the first increase in positive cases was around August or September 2020 (first wave), the second increase was in July 2021 (second wave), and the last wave came in January 2022. This shows that most countries in this area were affected by COVID-19 waves at about the same time. It was observed for most countries, the 2nd wave was deadlier than the other two.

### *F. Performance of Prediction Models*

Table I shows the performance score of algorithms used in work. We used the training and validation methods to check the training performance score of the algorithm after 782 days of data. We used the $k$-fold cross-validation method to test the models. This enables us to consider every observation for both training and testing in different folds to get a better estimate of how accurately the model will perform on new data. For $k$, we used the value 10, trained the model on 90% of the data, and then tested the model on the remaining 10%. We did this ten times, each part being a test sample once. Among all the algorithms, the Time Series Forecasting was the most accurate according to the regression model performance score.

TABLE I: Performance of Different Regression Models

| **Regression Algorithm** | **Performance** |
|---|---|
| Linear Regression | 30.3 |
| Polynomial Regression | 60.1 |
| KNN Regression | 87.8 |
| Time Series Forecasting Regression | 92.4 |

TABLE II: Time Series Forecasting Prediction

| **Prediction using Time Series Forecasting** | | | | |
|---|---|---|---|---|
| **Country Name** | **New Cases** | **New Deaths** | **New Tests** | **New Vaccinations** |
| Afghanistan | 382 | 4 | N/A | 11174 |
| Bangladesh | 8946 | 22 | 37891 | 996539 |
| Bhutan | 178 | 1 | 8509 | 9255 |
| India | 191570 | 693 | 1606488 | 5649273 |
| Maldives | 1792 | 1 | 5966 | 2330 |
| Nepal | 4159 | 9 | 11191 | 254171 |
| Pakistan | 5482 | 29 | 56915 | 1037582 |
| Sri Lanka | 1085 | 21 | 12448 | 66382 |

This could be because the Time Series Forecasting model was able to capture the temporal patterns and trends in the data compared to the others.

For each model, we split the data into training and testing sets and fitted the model on the training set using the optimal parameters. Then, we used the fitted model to predict the future values of the response variable on the testing set and evaluated the prediction accuracy using the Mean Absolute Error (MAE) metric. Then the final performance score was calculated by taking the mean across all the eight countries.

As shown in Table II, the Time Series Forecasting model predicts that COVID-19 cases and deaths will decline significantly in every nation in the next few days, possibly due to the increase in vaccinations. India has the highest mortality rate among the nations, which is mostly due to its large population size. Bangladesh, Pakistan, and Sri Lanka expect to see fewer deaths in the upcoming days than the current predictions. Maldives, Nepal, and Bhutan may be able to control the problem soon as they have high vaccination rates.

## V. Conclusion

This paper analyzes COVID-19 third wave in SAARC countries and forecasts the future situation using ML-based algorithms. We use data from various sources to visualize and compare the cases, tests, deaths, and vaccinations in each country. We use linear regression, polynomial regression, KNN regression, and Time Series Forecasting to predict the next 90 days based on 782 days of data. Time Series Forecasting is the most accurate according to the regression model performance score. We discuss the limitations of our study, such as data scarcity, virus mutation, and vaccination effect. Our findings show that COVID-19 has affected SAARC countries differently, depending on their population, healthcare, pandemic management, and vaccination rate. The second wave was the deadliest, and the third wave was the most contagious in this region. However, the predictions may not account for future variants that may be more resistant or lethal than the previous ones. Therefore, we suggest that SAARC countries monitor the situation closely and take preventive measures to avoid another wave of COVID-19.

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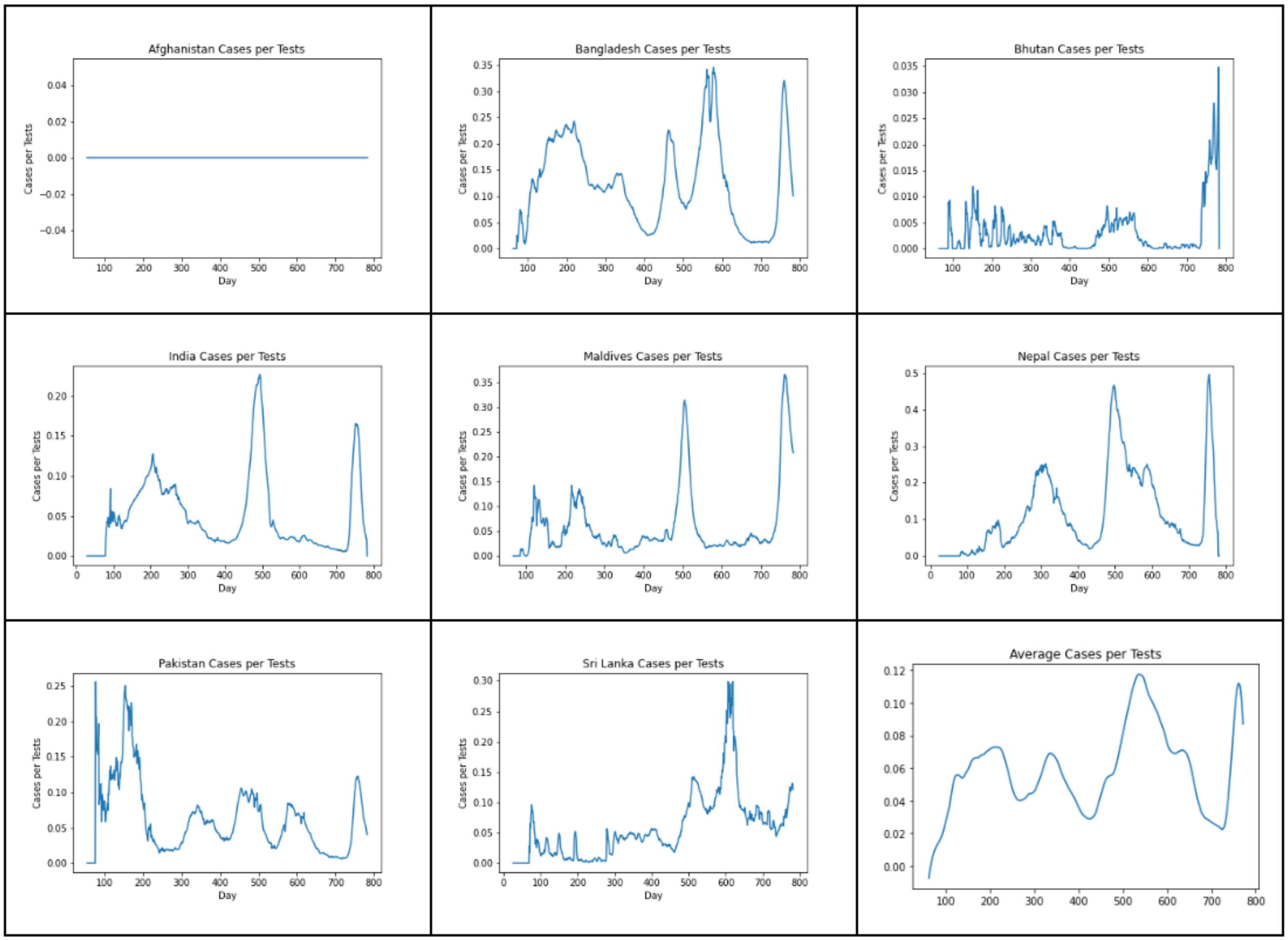


Fig. 2: Cases per Tests in the Countries and their average comparison


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